

Enterprise Predictive Analytics Engine

End-to-End Modern Data Stack Implementation Blueprint (SQL + DuckDB + Python + Streamlit)

1. Master Execution Roadmap & Target Architecture

This master implementation blueprint provides the technical roadmap to build the **Modern Data Stack (MDS)** architecture from scratch. The platform ingests 113,425 e-commerce transactions across 9 relational tables into **DuckDB**, executes 5 modular SQL models, trains 3 machine learning algorithms, and deploys an 8-page decision-support dashboard in **Streamlit**.

Directory / Module	Key Files & Artifacts	Architectural Responsibility
sql/	01_schema_and_views.sql to 05_monthly_kpis_mom.sql	Relational joins, window deduplication (ROW_NUMBER), RFM CTE modeling, MoM growth (LAG).
scripts/	run_sql_pipeline.py, train_models.py	Automated DuckDB SQL executor and ML model training/serialization pipeline.
models/	churn_risk_classifier.pkl, kmeans.pkl	Trained Random Forest classifier, KMeans (k=4) scaler, and Prophet metadata.
dashboards/	app.py, loader.py, pages/01-08	Multi-page Streamlit BI UI, Parquet caching gateway, real-time risk simulator.
tests/	test_sql_pipeline.py, test_models.py (7 files)	Automated Pytest suite verifying SQL queries, math formulas, and ML predictions.

2. The 5 Core SQL Analytics Models (sql/ Directory)

- **01_schema_and_views.sql**: Creates base relational view joining orders, items, payments, customers, products, and categories.
- **02_order_deduplication.sql**: Uses ROW_NUMBER() OVER (PARTITION BY order_id) to eliminate multi-item revenue double-counting.
- **03_rfm_segmentation.sql**: Multi-CTE query calculating Recency, Frequency, and Monetary spend grouped by customer_unique_id.
- **04_delivery_risk.sql**: Computes delivery delay vs SLA (DATEDIFF) and generates the binary low_review target for ML.
- **05_monthly_kpis_mom.sql**: Calculates monthly gross revenue, order volume, AOV, and MoM growth rate using LAG() OVER (ORDER BY month).

```
-- Key Model: 02_order_deduplication.sql (Window Function to prevent double-counting)
CREATE OR REPLACE VIEW v_orders_deduplicated AS WITH ranked_orders AS (
  SELECT order_id, customer_unique_id, purchase_timestamp, delivered_timestamp,
    estimated_delivery_timestamp, review_score, ROW_NUMBER() OVER (PARTITION BY order_id ORDER BY order_item_id) AS rn
  FROM v_master_transactions )
SELECT r.*, p.total_payment_value FROM ranked_orders r JOIN (
  SELECT order_id, SUM(payment_value) AS total_payment_value FROM olist_order_payments GROUP BY order_id) p ON r.order_id = p.order_id WHERE r.rn = 1;
```

3. Python Pipeline & Machine Learning Engines

The Python pipeline executes DuckDB transformations, trains three distinct machine learning tracks, and serializes model artifacts:

ML Track	Algorithm & Features	Production Benchmarks & Findings
Track 1: Customer Segmentation	StandardScaler + KMeans (k=4) Features: Recency, Frequency, Monetary	96.8% single-purchase churn rate (55.5% Recent, 41.3% Lapsed). Loyal repeat buyers (3.1%) spend 2.1x more (R\$ 428 vs R\$ 198).
Track 2: Dissatisfaction Classifier	Random Forest (n=200, depth=10) Features: Delays, Transit, Price, Freight, Categories	83.5% Accuracy, 0.430 F1 Score , 43.4% Precision. Delivery delays and transit time drive 76.7% of all bad reviews .
Track 3: Revenue Forecaster	Prophet Additive Time Series (90% CI) 11 Multi-Grain Cuts (Total, Categories, States)	Forecasts +19.4% revenue growth over 6 months (R\$ 1.66M to R\$ 1.98M/mo) with upper/lower bounds.

4. Streamlit BI Dashboard Architecture (8 Pages)

- **01_Executive_Overview.py**: Macro business KPIs (Sales, Orders, AOV, CSAT, On-Time Rate), monthly financial trends, geographic maps.
- **02_Customer_Analytics.py**: Cohort retention heatmaps, customer lifetime value distributions, repurchase timing interval analysis.
- **03_Customer_Risk.py**: Real-time what-if order risk simulator, dissatisfaction distribution, high-risk proactive intervention feed.
- **04_Revenue_Forecast.py**: 6-month Prophet revenue projections with 90% confidence bands, category and state dropdown selectors.
- **05_Model_Performance.py**: ROC-AUC curves, Precision-Recall curves, Confusion Matrices, and Feature Importance bar charts.
- **06_Data_Explorer.py**: Interactive SQL/Table query interface with dynamic multi-column filtering and instant CSV/JSON exports.
- **07_Customer_Segmentation.py**: 3D interactive RFM cluster scatter plot, segment profile comparison cards, and marketing playbooks.
- **08_About.py**: Architecture documentation, pipeline flowcharts, dataset schema dictionary, and deployment guidelines.

5. Automated Quality Assurance & Execution Checklist

Step / Module	Command Line Instruction	Verification Criteria
1. Dependency Setup	pip install -r requirements.txt	DuckDB, Streamlit, scikit-learn, Prophet installed cleanly.
2. SQL Pipeline	python scripts/run_sql_pipeline.py	DuckDB database created; clean Parquet files exported.
3. ML Training	python scripts/train_models.py	Random Forest (F1=0.43), KMeans (k=4), Prophet models serialized.
4. Automated Tests	pytest -v	All 7 test suites pass (SQL parity, KPIs, models, loaders).
5. Launch UI	streamlit run app.py	Interactive 8-page dashboard renders at http://localhost:8501.